

Lab_13_Report

German Vega

Pathway Analysis from RNA-Seq Results

Overview Analysis of high-throughput biological data typically yields a list of genes or proteins requiring further interpretation - for example the ranked lists of differentially expressed genes we have been generating from our RNA-seq analysis to date.

###Section 1. Differential Expression Analysis

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Loading required package: generics
```

```
Attaching package: 'generics'
```

```
The following objects are masked from 'package:base':
```

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

```
Attaching package: 'BiocGenerics'
```

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

```
colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars
```

Loading required package: Biobase

Welcome to Bioconductor

```
Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.
```

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

```
rowMedians
```

The following objects are masked from 'package:matrixStats':

```
anyMissing, rowMedians
```

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
```

```
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
      condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv(countFile, row.names=1)
head(countData)
```

```
      length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092    918         0         0         0         0         0
ENSG00000279928    718         0         0         0         0         0
ENSG00000279457   1982        23        28        29        29        28
ENSG00000278566    939         0         0         0         0         0
ENSG00000273547    939         0         0         0         0         0
ENSG00000187634   3214       124       123       205       207       212
      SRR493371
ENSG00000186092         0
ENSG00000279928         0
ENSG00000279457        46
ENSG00000278566         0
ENSG00000273547         0
ENSG00000187634       258
```

Q1. Complete the code below to remove the troublesome first column from countData

```
# Note we need to remove the odd first $length col
countData <- as.matrix(countData[,-1])
head(countData)
```

```
      SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092         0         0         0         0         0         0
ENSG00000279928         0         0         0         0         0         0
ENSG00000279457        23        28        29        29        28        46
```

ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will rowSums() of countData return and how could you use it in this context?

```
# Filter count data where you have 0 read count across all samples.
countData = countData[rowSums(countData) > 1, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
dds = DESeqDataSetFromMatrix(countData=countData,
                              colData=colData,
                              design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```
class: DESeqDataSet
dim: 15280 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15280): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor
```

```
res = results(dds)
```

Q3. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

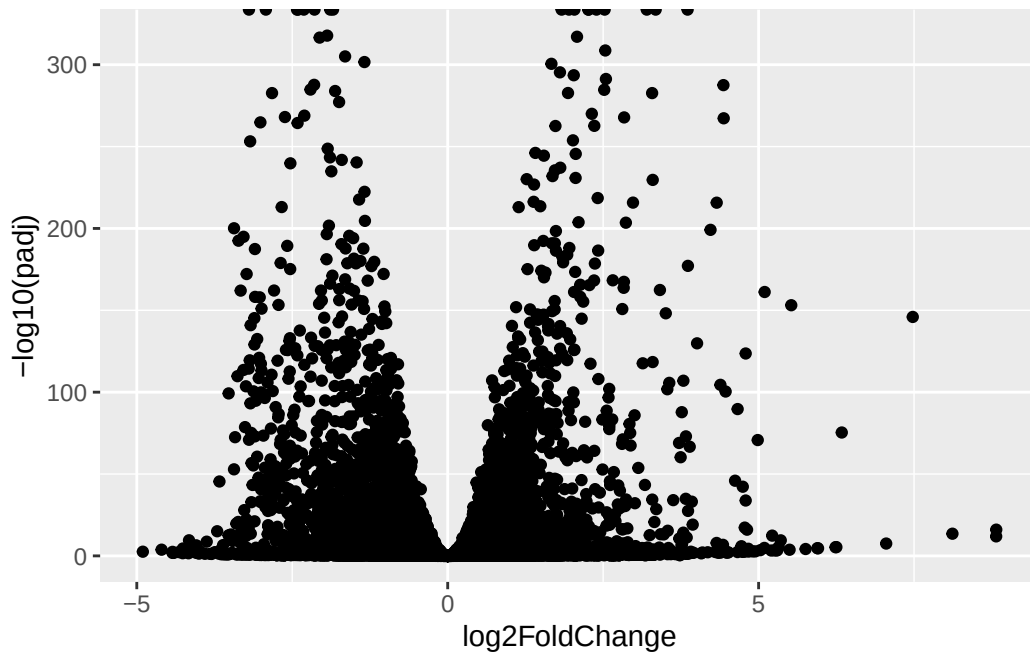
```
out of 15280 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)           : 4351, 28%
LFC < 0 (down)         : 4399, 29%
outliers [1]           : 0, 0%
low counts [2]         : 590, 3.9%
(mean count < 1)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

###Volcano Plot

```
library(ggplot2)

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange,
       y = -log10(padj)) +
  geom_point()
```

Warning: Removed 590 rows containing missing values or values outside the scale range (``geom_point()``).



Q4. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:

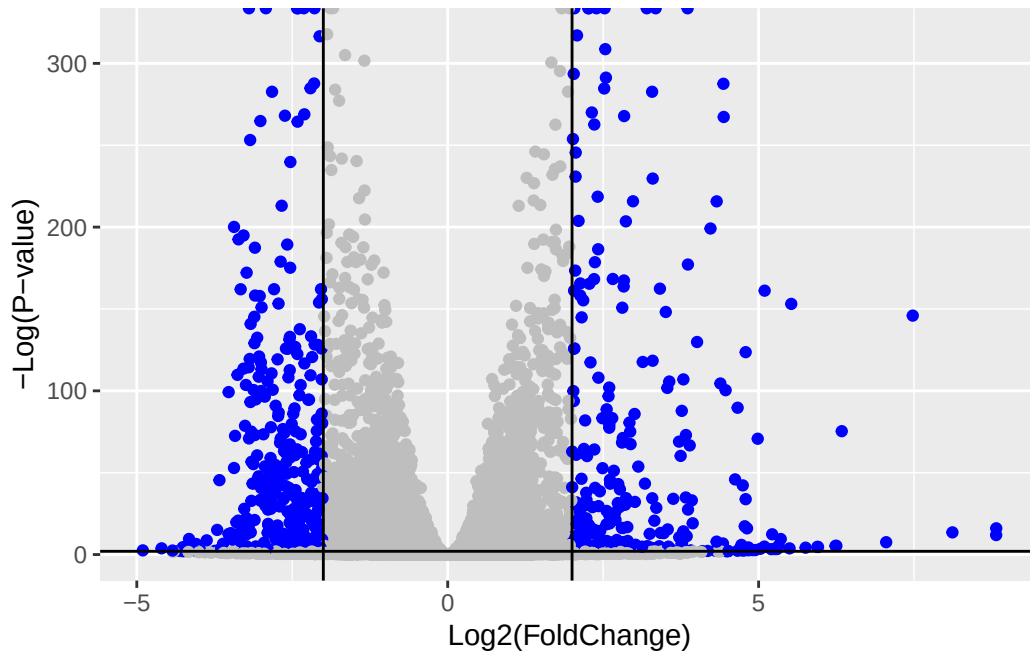
```
# Make a color vector for all genes
mycols <- rep("gray", nrow(res))

# Color blue the genes with fold change above 2
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

# Color gray those with adjusted p-value more than 0.01
mycols[ res$padj > 0.01 ] <- "gray"

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange,
      y = -log10(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = -log10(0.01))
```

Warning: Removed 590 rows containing missing values or values outside the scale range (``geom_point()``).



###Adding gene annotation

Q5. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"       "ENSEMBL"     "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"      "EVIDENCE"    "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"    "GO"          "GOALL"       "IPI"          "MAP"
[16] "OMIM"        "ONTOLOGY"    "ONTOLOGYALL" "PATH"         "PFAM"
[21] "PMID"        "PROSITE"     "REFSEQ"      "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```



```
res$symbol = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="SYMBOL",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
                    keys=row.names(res),
                    keytype="ENSEMBL",
                    column="ENTREZID",
                    multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
                  keys=row.names(res),
                  keytype="ENSEMBL",
                  column="GENENAME",
                  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.9221	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.7169	0.2573837	0.1027266	2.505522	1.22271e-02

ENSG00000188157	9128.4394	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000131591	156.4791	0.1965923	0.1456109	1.350121	1.76977e-01
	padj	symbol	entrez		name
	<numeric>	<character>	<character>		<character>
ENSG00000279457	6.85033e-01	NA	NA		NA
ENSG00000187634	5.14038e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.75978e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13044e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19159e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.02066e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30113e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.36679e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.20596e-16	AGRN	375790		agrin
ENSG00000131591	2.60893e-01	C1orf159	54991	chromosome 1 open re..	

Q6. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$pvalue),]
write.csv(res, file="deseq_results.csv")
```

###Section 2. Pathway Analysis

```
# Run in your R console (i.e. not your Rmarkdown doc!)
BiocManager::install( c("pathview", "gage", "gageData") )
```

Bioconductor version 3.23 (BiocManager 1.30.27), R 4.6.0 (2026-04-24)

Warning: package(s) not installed when version(s) same as or greater than current; use
`force = TRUE` to re-install: 'pathview' 'gage' 'gageData'

Warning in .available.both(repos, method, ...): Some listed binary packages
have no source

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
```

formally cite the original Pathview paper (not just mention it) in publications or products. For details, do `citation("pathview")` within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

#####

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
```

```

[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"

```

```

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)

```

```

      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288

```

```

# Get the results
keggres = gage(foldchanges, gsets=kegg.sets.hs)

```

```

attributes(keggres)

```

```

$names
[1] "greater" "less" "stats"

```

```

# Look at the first few down (less) pathways
head(keggres$less)

```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	1.003993e-05	-4.353454	1.003993e-05
hsa03030 DNA replication	8.909558e-05	-3.968611	8.909558e-05
hsa03013 RNA transport	1.470985e-03	-3.007794	1.470985e-03
hsa04114 Oocyte meiosis	1.946905e-03	-2.921710	1.946905e-03
hsa03440 Homologous recombination	2.941989e-03	-2.868141	2.941989e-03

hsa00010	Glycolysis / Gluconeogenesis	6.059196e-03	-2.558327	6.059196e-03
		q.val	set.size	exp1
hsa04110	Cell cycle	0.001606390	120	1.003993e-05
hsa03030	DNA replication	0.007127646	36	8.909558e-05
hsa03013	RNA transport	0.077876200	143	1.470985e-03
hsa04114	Oocyte meiosis	0.077876200	99	1.946905e-03
hsa03440	Homologous recombination	0.094143663	28	2.941989e-03
hsa00010	Glycolysis / Gluconeogenesis	0.161578552	48	6.059196e-03

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs/143/Proect_Lab_13

Info: Writing image file hsa04110.pathview.png

```
# A different PDF based output of the same data
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

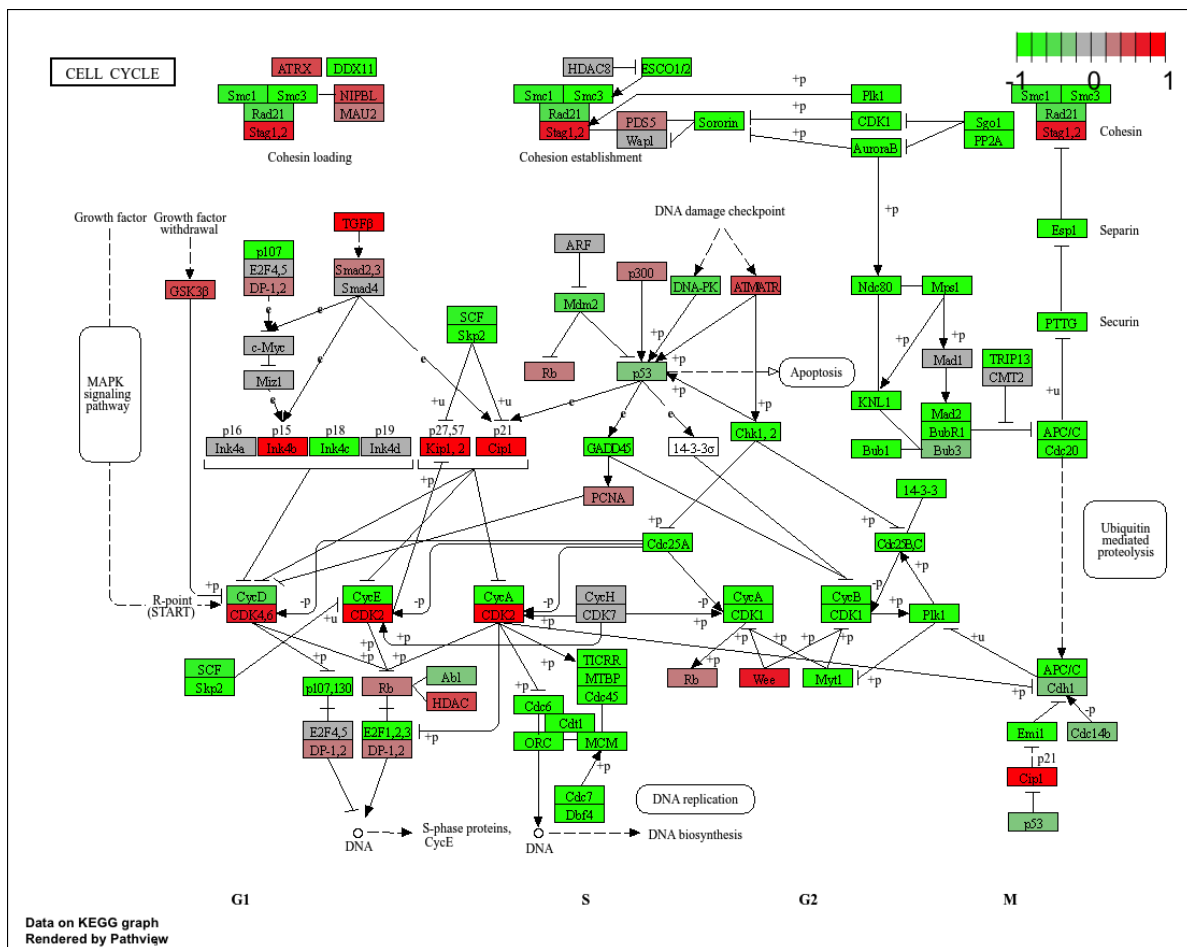
Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]
[1,] "9"  "300"
[2,] "9"  "306"
```

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs/143/Proect_Lab_13

Info: Writing image file hsa04110.pathview.pdf

Warning in text.default(xloc, yloc, label = labs[sub2grp[sub.idx, 1]], cex = cex): length(lab
'labels' truncated to length 36



Focus on top 5 upregulated pathways here for demo purposes only

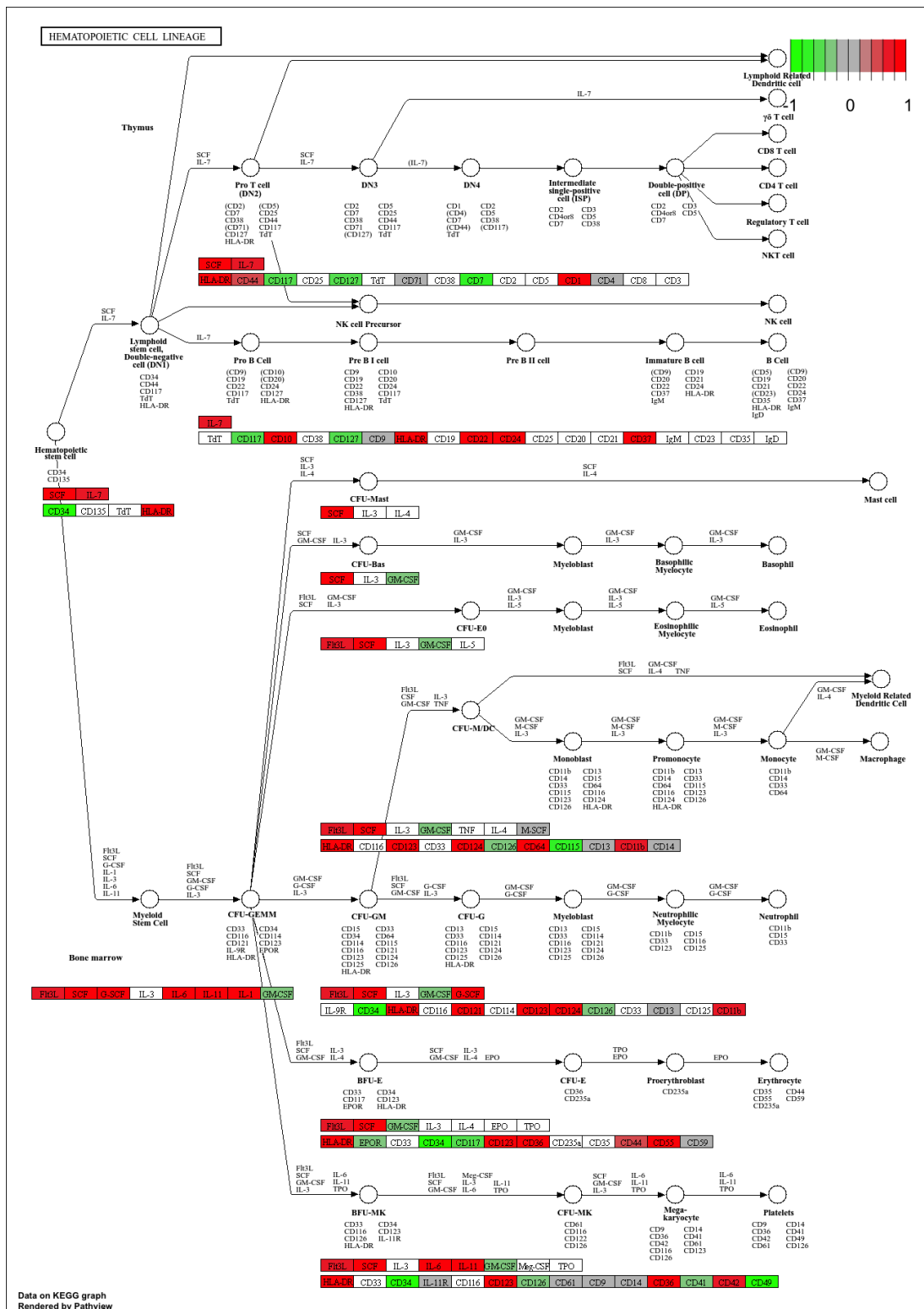
```
keggrespathways <- rownames(keggres$greater)[1:5]
```

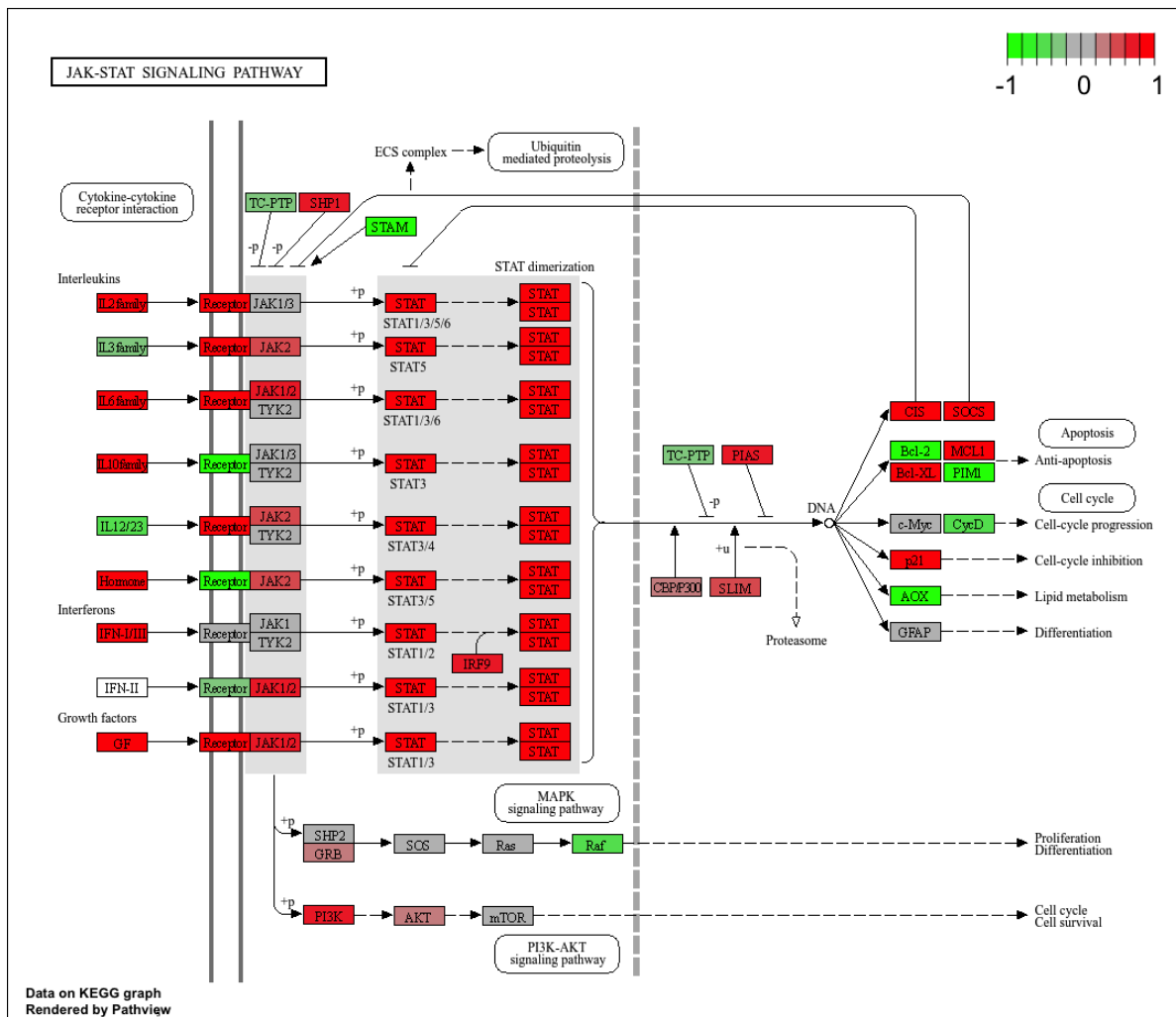
```
# Extract the 8 character long IDs part of each string
```

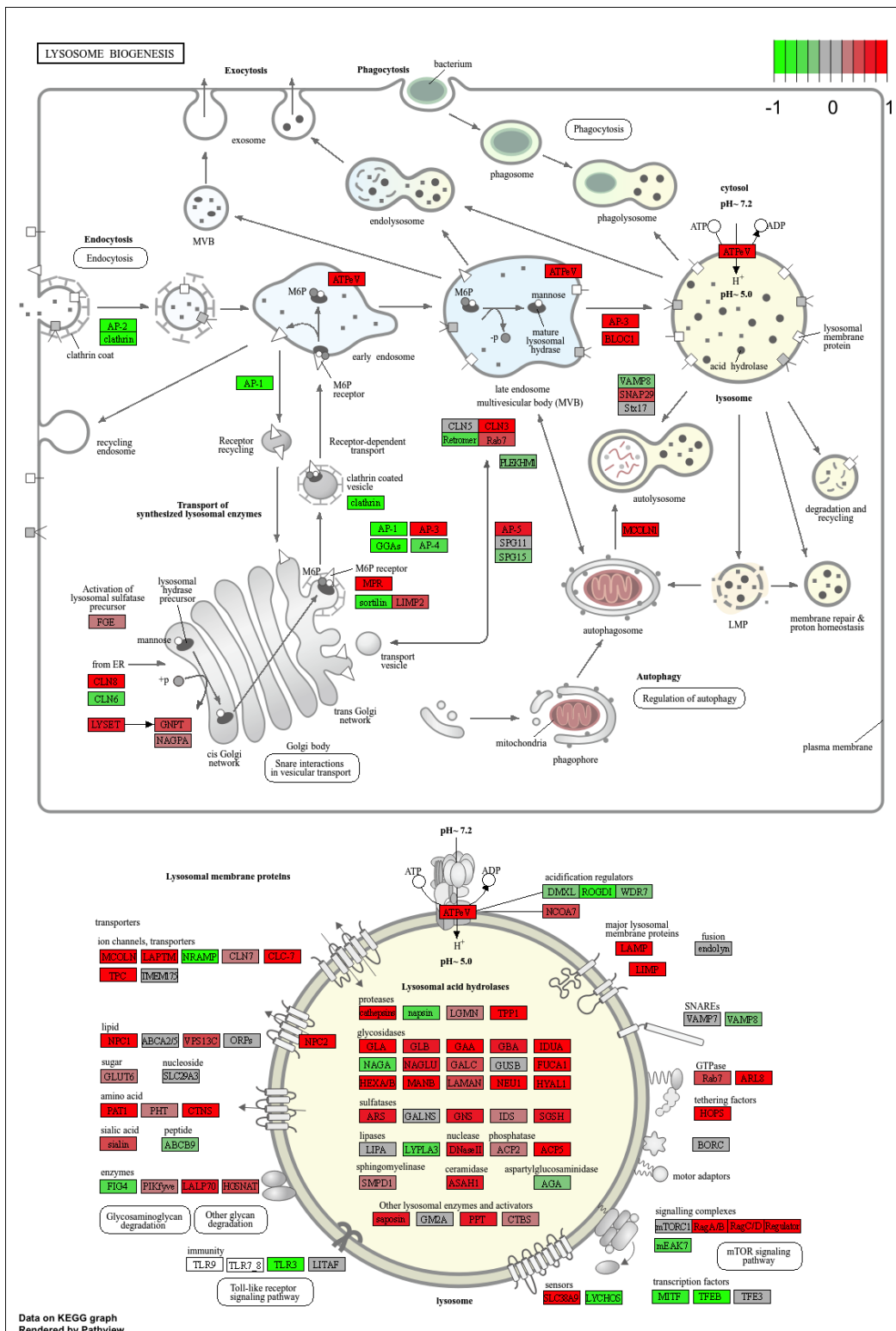
```
keggresids = substr(keggrespathways, start=1, stop=8)
```

```
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa04142" "hsa00140" "hsa04740"
```







Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa00140.pathview.png

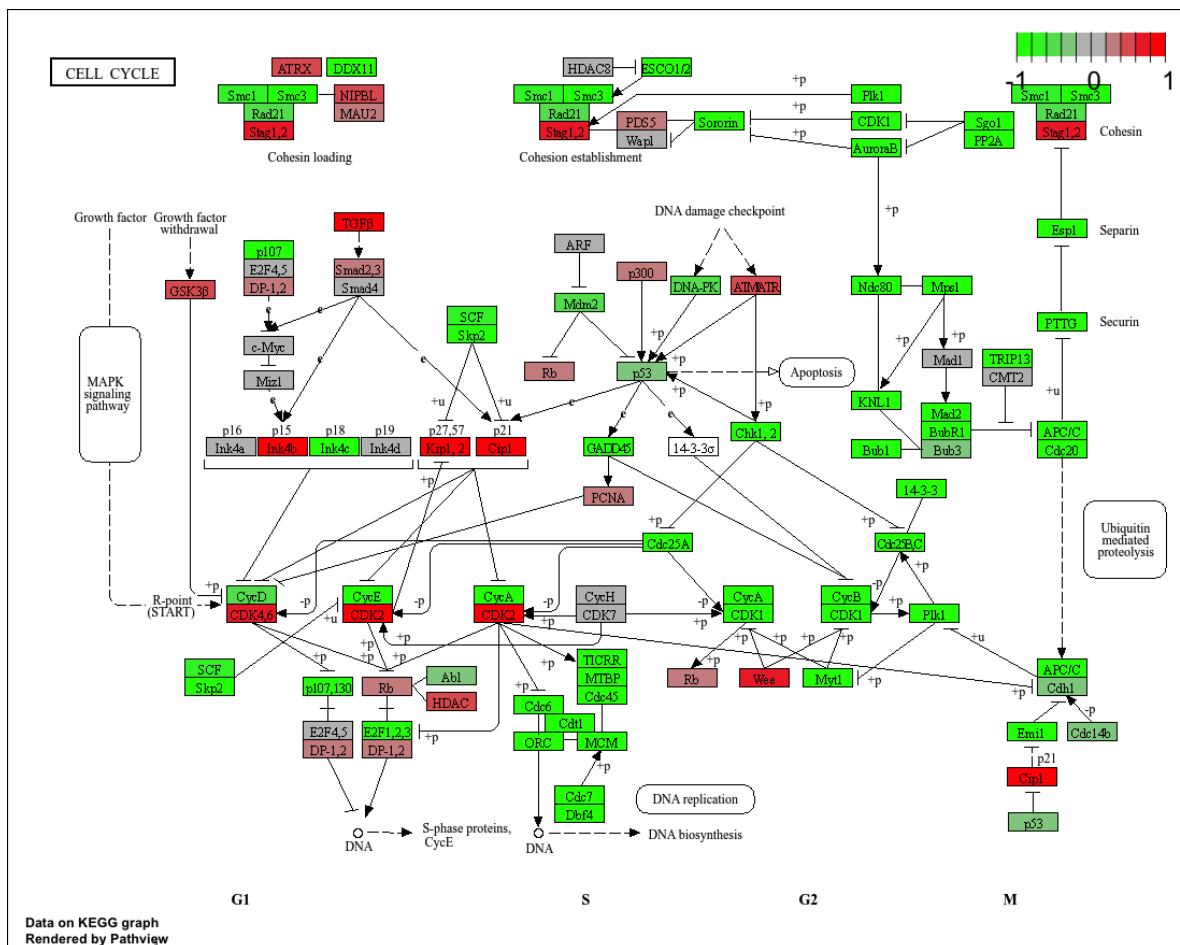
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa04740.pathview.png

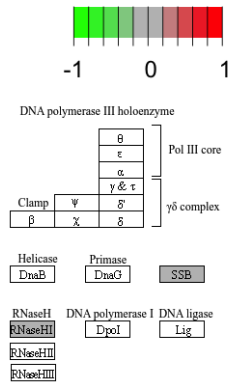
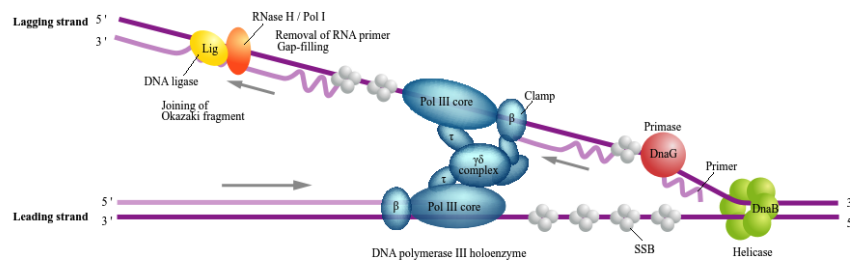
Q7. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

A: Same procedure as before, just swap *keggresgreater* for *keggresless* to pull the top 5 downregulated pathways, extract the 8-character IDs with `substr()`, and pass them into `pathview()` with the same `gene.data` and `species="hsa"` arguments.

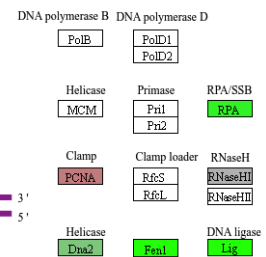
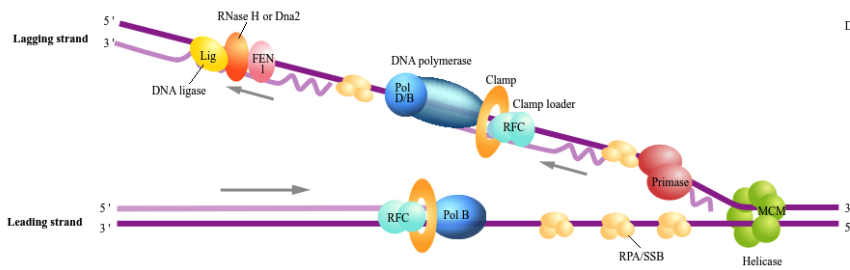


DNA REPLICATION

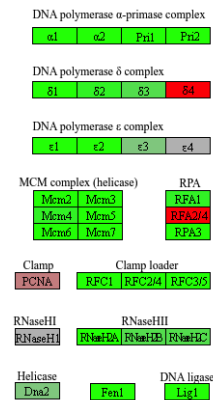
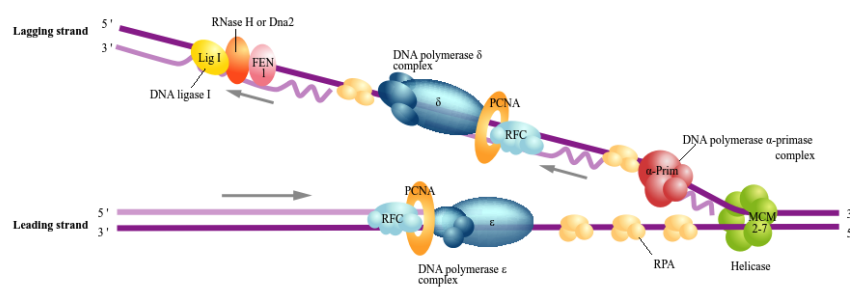
Replication complex (Bacteria)



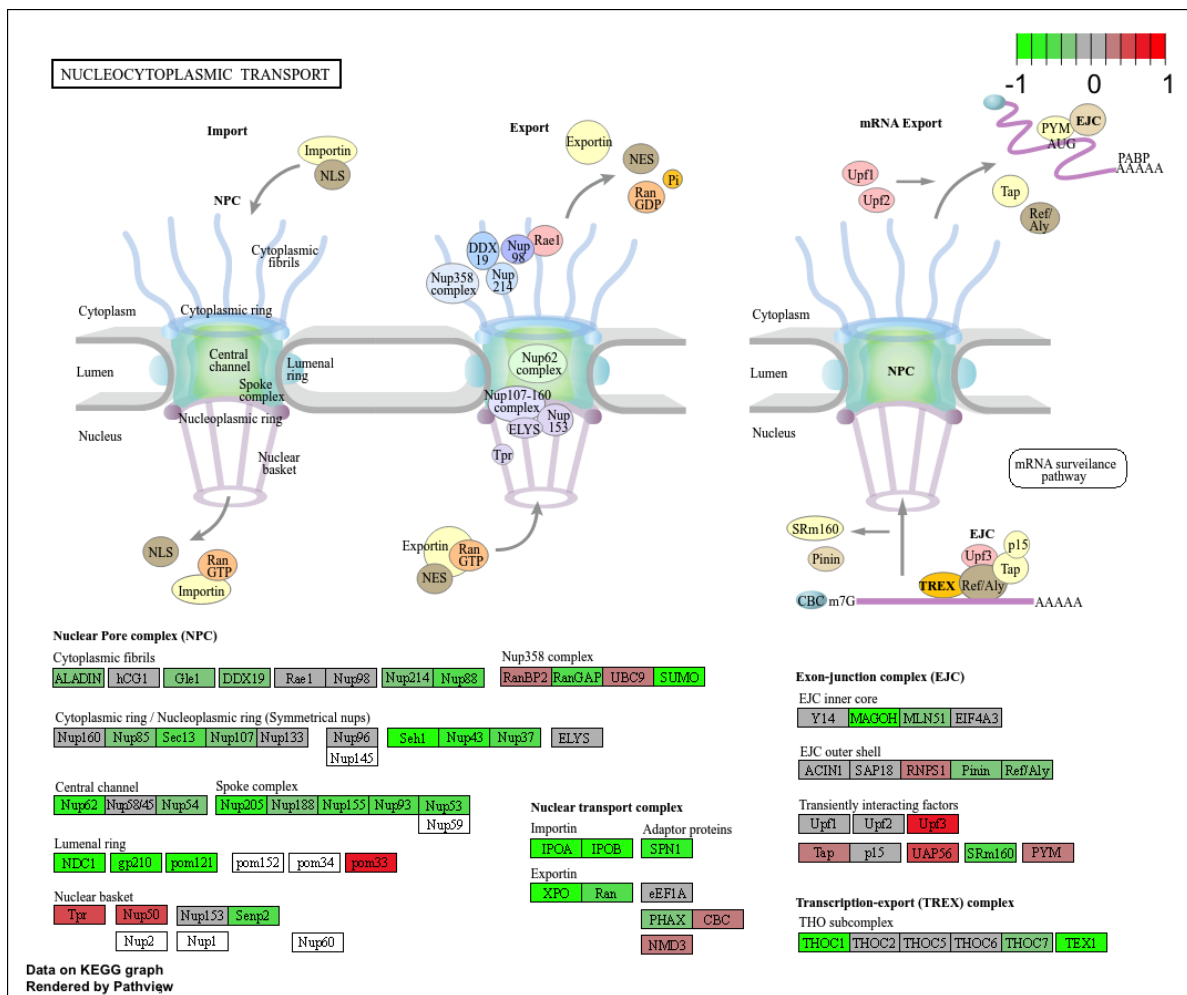
Replication complex (Archaea)

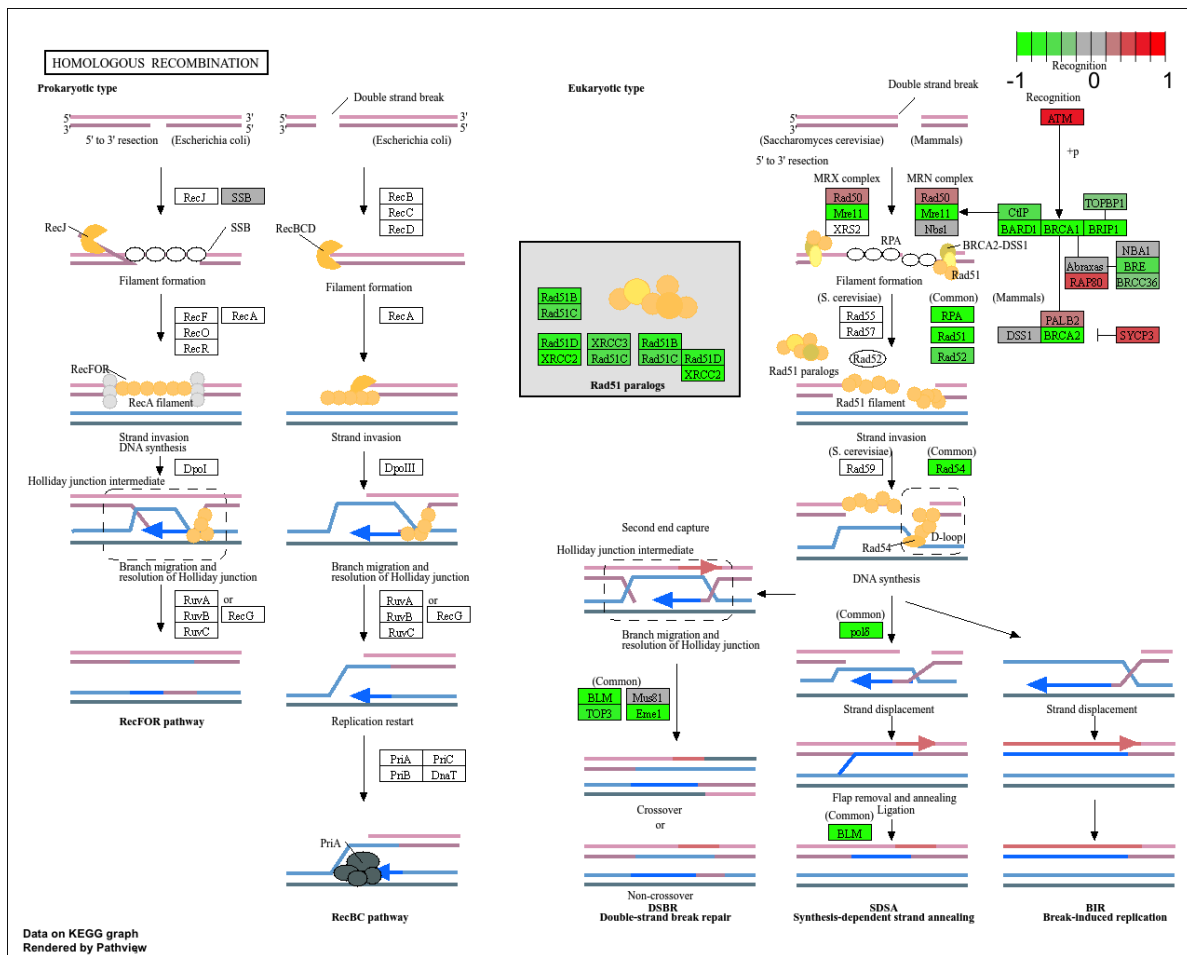


Replication complex (Eukaryotes)



Data on KEGG graph
Rendered by Pathview





Focus on top 5 downregulated pathways

```
keggrespathways_down <- rownames(keggres$less)[1:5]
```

Extract the 8 character long IDs

```
keggresids_down <- substr(keggrespathways_down, start=1, stop=8)
```

```
keggresids_down
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

Plot pathview for top 5 downregulated pathways

```
pathview(gene.data=foldchanges, pathway.id=keggresids_down, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa04114.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/germanvega_25/Library/Mobile Documents/com~apple~CloudDocs, 143/Proect_Lab_13

Info: Writing image file hsa03440.pathview.png

###Section 3. Gene Ontology (GO)

```

data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)

```

\$greater

	p.geomean	stat.mean	p.val
GO:0007156 homophilic cell adhesion	4.892477e-05	3.971899	4.892477e-05
GO:0060429 epithelium development	6.727546e-05	3.834595	6.727546e-05
GO:0007610 behavior	1.735208e-04	3.594057	1.735208e-04
GO:0048729 tissue morphogenesis	2.470962e-04	3.498983	2.470962e-04
GO:0002009 morphogenesis of an epithelium	3.227439e-04	3.429317	3.227439e-04
GO:0016337 cell-cell adhesion	8.195506e-04	3.163057	8.195506e-04
	q.val	set.size	expl
GO:0007156 homophilic cell adhesion	0.1337100	107	4.892477e-05
GO:0060429 epithelium development	0.1337100	478	6.727546e-05
GO:0007610 behavior	0.2299150	402	1.735208e-04
GO:0048729 tissue morphogenesis	0.2455519	403	2.470962e-04
GO:0002009 morphogenesis of an epithelium	0.2565814	326	3.227439e-04
GO:0016337 cell-cell adhesion	0.3656453	318	8.195506e-04

\$less

	p.geomean	stat.mean	p.val
GO:0000279 M phase	1.475361e-16	-8.323749	1.475361e-16
GO:0048285 organelle fission	7.498412e-16	-8.160305	7.498412e-16
GO:0000280 nuclear division	2.135098e-15	-8.034814	2.135098e-15
GO:0007067 mitosis	2.135098e-15	-8.034814	2.135098e-15
GO:0000087 M phase of mitotic cell cycle	5.927567e-15	-7.891758	5.927567e-15
GO:0007059 chromosome segregation	1.055918e-11	-6.988373	1.055918e-11
	q.val	set.size	expl
GO:0000279 M phase	5.864560e-13	492	1.475361e-16
GO:0048285 organelle fission	1.490309e-12	373	7.498412e-16
GO:0000280 nuclear division	2.121754e-12	349	2.135098e-15
GO:0007067 mitosis	2.121754e-12	349	2.135098e-15
GO:0000087 M phase of mitotic cell cycle	4.712416e-12	359	5.927567e-15
GO:0007059 chromosome segregation	6.995458e-09	141	1.055918e-11

```
$stats
```

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	3.971899	3.971899
G0:0060429 epithelium development	3.834595	3.834595
G0:0007610 behavior	3.594057	3.594057
G0:0048729 tissue morphogenesis	3.498983	3.498983
G0:0002009 morphogenesis of an epithelium	3.429317	3.429317
G0:0016337 cell-cell adhesion	3.163057	3.163057

```
###Section 4. Reactome Analysis
```

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8149"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q8: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

A: The hsa04110 Cell cycle has the most significant p-value (8.99e-06) in the KEGG results, which generally matches what Reactome returns for this dataset. Minor differences between the two are expected since KEGG and Reactome curate pathways independently, and apply different statistical methods, KEGG via gage uses a directional t-test across all genes while Reactome uses over-representation analysis on a filtered gene list, so rankings can shift even when the underlying biology is the same.